INCH POUND
MIL-P-70825A (AR)
2 June 1994
SUPERSEDING
MIL-P-70825 (AR)
09 September 1988

MILITARY SPECIFICATION

PRIMER, BATTERY, STAB/ELECTRIC PA536 PARTS FOR, AND LOADING, ASSEMBLING AND PACKING

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the parts for, and the loading, assembling and packing for one type of primer designated as Primer, Battery, Stab/Electric, PA536.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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FEDERAL

A-A-208 - Ink, Marking, Stencil, Opaque (Porous and Non-Porous Surfaces)

MILITARY

MIL-P-116 - Preservation, Methods of
MIL-B-117 - Bag, Sleeve and Tubing - Interior Packaging
MIL-G-45204 - Gold Plating, Electrodeposited
MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-202 - Electronic and Electrical Component Parts, Test Methods for
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1190 - Minimum Guidelines for Level C Preservation, Packing and Marking

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from DODSSP- Customer Service, Standardization Documents Order Desk, 700 Robbins Avenue, Bldg. 4D, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (see 6.5)

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

Product and Test Fixture Drawings

8796522 - Marking Diagram and Sealing for Wood Packing Boxes
12550754 - Test Sleeve

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12550991 - Battery Primer Assembly, Stab/Electrical, PA536

Inspection Equipment Drawings

81-3-150 - Fixture, Testing Detonator Sensitivity Assembly Complete

(Copies of other Government documents, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

PUBLICATIONS

CODE OF FEDERAL REGULATIONS

Title 49 - Transportation, Parts 100-199

(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. Orders should specify, "49 CFR 100-199 (latest revision)".)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained. (See contract provisions for additional precedence criteria).

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The Components and assemblies shall comply with all requirements specified on Drawing 12550991, all associated drawings, and with all requirements specified in applicable specifications and standards.

3.3 Moisture content.

3.3.1 Primer input mixture. The moisture content of the primer input mixture at the time of loading the primer shall not exceed 0.30 percent when tested as specified in 4.5.1.

3.4 Functioning tests-sensitivity and output.

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3.4.1 No-fire electrical test. The Battery Primer assembly shall not function when a direct current of 10.0 ± .5 milliamperes is applied for a minimum of 10 seconds, in accordance with 4.5.2.1.

3.4.2 No-fire mechanical test. The Battery Primer assembly shall not function when dropped from a height of 5 feet minimum onto a flat steel plate, in accordance with 4.5.2.3.

3.4.3 All-fire output test. The Battery Primer shall function on the first attempt and shall expand the diameter of the witness sleeve by not less than 0.035 inch when tested as specified in 4.5.2.2.

3.4.4 All-fire.

3.4.4.1 Electrical all-fire function. The PA536 after being conditioned at 70 degrees ± 10 degrees F for two hours minimum shall function on the first attempt when energy from the circuit shown in Figure 1 is applied.

3.4.4.2 Mechanical all-fire function. The Battery Primer assembly, after being conditioned at 70 degrees ± 10 degrees F for two hours minimum shall function when a maximum of 5 inch-ounces of energy is applied to a steel firing pin with a needle having a 16 degrees -2 degree right circular cone and a .007 ± .001 inch flat. The firing pin shall be positioned within a .07 inch diametral area on the stab sensitive surface.

3.5 Environmental tests.

3.5.1 Resistance between pin and cup, thermal shock, vibration and function. The Battery Primer shall comply with the resistance requirements of Drawing 12550991 and with the requirements of 3.4.1 after being exposed to thermal shock and vibration as specified in 4.5.3.1.

3.5.2 Hot temperature.

3.5.2.1 Electrical all-fire. The PA536 shall comply with the electric functional requirements of 3.4.4.1 after being conditioned at 160 degrees ± 5 degrees F for two hours minimum and tested as specified in 4.5.2.4.1.

3.5.2.2 Mechanical all-fire. The PA536 shall comply with the mechanical functional requirements of 3.4.4.2 after being conditioned at 160 degrees ± 5 degrees F for two hours minimum and tested as specified in 4.5.2.4.2.

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NOTES:
1. TOLERANCES ON ELECTRICAL PARTS ARE ± 10% UNLESS OTHERWISE NOTED.

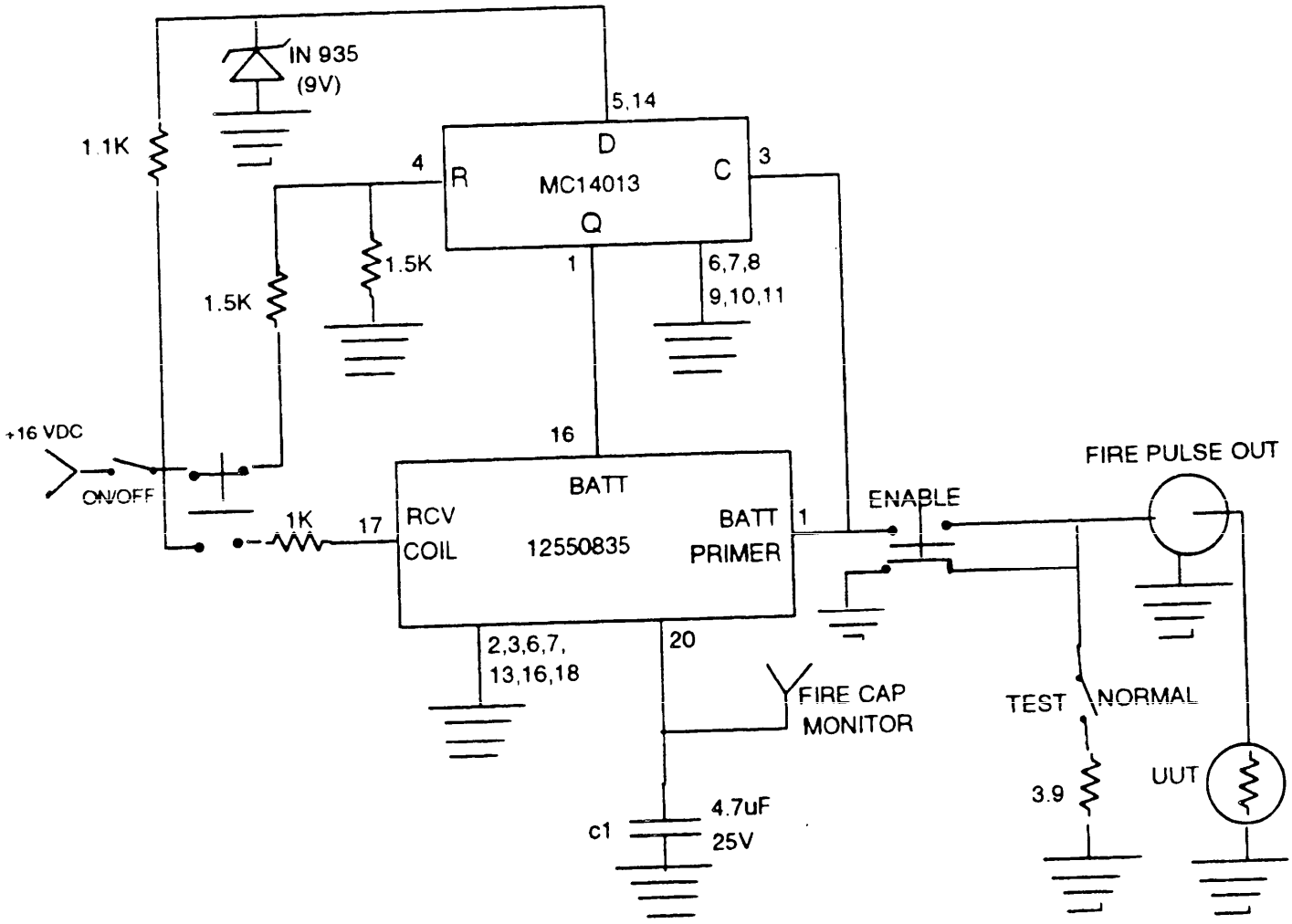


FIGURE 1. Battery primer initiator circuit

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3.5.3 Cold temperature.

3.5.3.1 Electrical all-fire. The PA536 shall comply with the electric functional requirements of 3.4.4.1 after being conditioned at -65 degrees ± 5 degrees F for two hours minimum and tested as specified in 4.5.2.4.1.

3.5.3.2 Mechanical all-fire. The PA536 shall comply with the mechanical functional requirements of 3.4.4.2 after being conditioned at -65 degrees ± 5 degrees F for two hours minimum and tested as specified in 4.5.2.4.2.

3.5.4 Waterproofness. The Battery Primer shall comply with the requirements of 3.4.4 after being submerged in water to a minimum (min.) depth of two (2) inches for not less than 48 hours, at a temperature of 70 degrees ± 10 degrees F, when tested as specified in 4.5.3.2.

3.6 First article. When specified in the contract or purchases order (see 6.2), a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.7 Change of qualified product. After first article, the manufacturer shall notify the purchaser prior to the implementation of any change which may affect performance, quality, reliability or interchangeability. Such notification shall include a thorough description of the proposed change and a suggested test plan designed to demonstrate that the change will not adversely affect performance, quality, reliability or interchangeability and that the changed product will continue to meet the specification requirement. The purchaser will review the proposed change and test plan and advise the manufacturer of the testing required to obtain approval of the proposed change. Upon completion of the prescribed tests, the results shall be submitted to the Purchaser for review. Based upon the review the purchaser will approve or disapprove the change and notify the manufacturer. At the manufacturer's option, devices incorporating the change may be manufactured and tested prior to approval however, all shipments of these changed devices shall be withheld until approval is granted by the purchaser. Changes representative of those which are subject to this requirement are any design changes in the manufacturing processing such as:

a. Package structure and material changes or changes to the geometry.

b. Changes to process flow charts, assembly flow chart and testing flow charts.

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c. Conductor materials.

d. Assembly operations move from one line or building to another.

3.8 Workmanship. All parts and assemblies shall be fabricated and loaded in a thorough, workmanlike manner. They shall be clean and free of burrs, chips, sharp edges, cracks, surface defects, dirt, grease, oil, rust, corrosion products, and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 General provisions. Unless otherwise specified herein, the provisions of MIL-A-48078 apply and form a part of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection

b. Quality conformance inspection

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the items as required per TABLE I in sample quantities as indicated.

PART DESCRIPTION	DRAWING	QUANTITY
Cup	12550792	10
Header Assembly	12550795	25
Bridged Header Assembly	12550794	25
Bridged Header and Ignition Assembly	12550793	25
Battery Primer Assembly	12550991	27[illegible]
Sleeve, Header	12550796	10

4.3.2 Inspection to be performed. See MIL-A-48078 and TABLE I specified herein.

4.3.3 Rejection. See MIL-A-48078.

4.4 Qualification conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of primers shall contain:

a. Parts bearing the same lot interfix number from one manufacturer.

b. Primer input mixture from not more than one lot.

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TABLE I - First article inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
	Battery primer assembly and components		See Below
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Cup (Dwg. 12550792) Examination for defects	10	3.2	4.4.2.1
	Sleeve, Header (Dwg 12550796)	10	3.2	4.4.2.8
	Header Assembly (Dwg. 12550795) Examination for defects	25	3.2	4.4.2.6
	Bridged Header Assembly (Dwg. 12550794) Examination for defects	25	3.2	4.4.2
	Bridged Header and Ignition Assembly (Dwg. 12550793) Examination for defects	25	3.2	4.4.2.5
	Battery Primer Assembly, Stab/ Electrical PA536 (Dwg 12550991) Examination for defects Bridgewire resistance test Sensitivity and output test	Note 1 Note 1 150	3.2 3.2 3.2	4.4.2.7 4.5.5 4.5.2

NOTES

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TABLE I - First article inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE: Battery primer assembly and components	SHEET 2 OF 2	DRAWING NUMBER: See Below
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Battery Primer Assembly, Stab/ Electrical, PA536 (Dwg 12550991) (Continued)			
	No-fire test - electrical	150-0-1	3.4.1	4.5.2.1
	- mechanical (after electrical)	150-0-1	3.4.2	4.5.2.3
	All-fire output test:	75-0-1	3.4.3	4.5.2.2
	All-fire function test - mechanical	75-0-1	3.4.4.2	4.5.2.4.2
	Environmental Tests	60	3.5	4.5.3
	Thermal shock, vibration and function			
	No-fire test - electrical	30-0-1	3.4.1	4.5.2.1
	- mechanical (after electrical)	30-0-1	3.4.2	4.5.2.3
	All-fire test - electrical hot	15-0-1	3.5.2.1	4.5.2.4.1
	- electrical cold	15-0-1	3.5.3.1	4.5.2.4.1
	- mechanical hot	15-0-1	3.5.2.2	4.5.2.4.2
	- mechanical cold	15-0-1	3.5.3.2	4.5.2.4.2
	Waterproofness test	60	3.5.4	4.5.3.2
	All-fire test - electrical hot	15-0-1	3.5.2.1	4.5.2.4.1
	- electrical cold	15-0-1	3.5.3.1	4.5.2.4.1
	- mechanical hot	15-0-1	3.5.2.2	4.5.2.4.2
	- mechanical cold	15-0-1	3.5.3.2	4.5.2.4.2

NOTES:

1. The entire quantity of battery primers required for the tests below shall be subjected to the examinations specified prior to functional testing.

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4.4.2 Examination and tests.

a. Classification of characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified. When cited herein, attributes sampling inspection shall be conducted in accordance with TABLE II below, using the inspection levels stated in the Classification of Characteristics paragraphs.

TABLE II Attributes sampling inspection

Lot Size	Inspection Levels					
	I	II	III	IV	V	VI
2 to 8	*	*	*	*	5	3
9 to 15	*	*	*	13	5	3
16 to 25	*	*	*	13	5	3
26 to 50	*	*	32	13	5	3
51 to 90	*	*	32	13	13	5
91 to 150	*	125	32	13	13	5
151 to 280	*	125	32	32	20	8
281 to 500	*	125	32	32	20	8
501 to 1200	*	125	80	50	20	13
1201 to 3200	1250	125	80	50	32	13
3201 to 10000	1250	125	125	50	32	13
10001 to 35000	1250	315	125	80	50	13
35001 to 150000	1250	315	125	80	50	13
150001 to 500000	1250	500	200	125	50	13
500001 and over	1250	500	200	125	50	13

Numbers under inspection levels indicate sample size; asterisks (*) indicate one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection. Accept on zero and reject on one or more failures/defects for all inspection levels.

b. Alternative quality conformance provisions. Unless otherwise specified herein or provided for in the contract, alternative quality conformance procedures, methods or equipment, such as statistical process control, tool control, other types of sampling plans, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions herein. Prior to applying such alternative procedures, methods or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation

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reactivity of the polymeric compound with ... specified in 3.3.1.10, the

effectiveness of each proposed alternative is equal to or better than the specified quality conformance provision(s) herein. In case of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or inspection system, as applicable.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Cup	1 OF 1	12550792
			NEXT HIGHER ASSEMBLY: 12550791

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Outside Diameter	Level IV	3.2	Gage
102	Overall length	Level IV	3.2	Gage
103	Thickness through closed end	Level IV	3.2	Gage
Minor				
201	Evidence of poor workmanship	Level V	3.8	Visual
202	Inside Diameter	Level V	3.2	Gage

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

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PARAGRAPH	TITLE		SHEET 1 OF 1	DRAWING NUMBER
4.4.2.2	Electrode			12550797
				NEXT HIGHER ASSEMBLY
				12550795
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major 101	Diameter of electrode	Level IV	3.2	Gage
Minor 201	Evidence of poor workmanship	Level V	3.8	Visual

NOTES:

AMSMC Form 1570b-E, 1 Jul 89

Replaces AMSMC Form 1570, 1 Feb 85, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Bridgewire	1 OF 1	9326203
			NEXT HIGHER ASSEMBLY: 12550794

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major 101	Moleculoy Wire Diameter	See Note	3.2	Gage
Minor 201	Evidence of poor workmanship	Level V	3.8	Visual

NOTES:
Five (5) measurements shall be made in the first two (2) feet of wire on each spool, and at least once per shift thereafter.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Charged cup assembly	1 OF 1	12550791
			NEXT HIGHER ASSEMBLY: 12550991

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Height of total charge	Level IV	3.2	Gage
102	Moisture content of primer input mixture	Note 1	3.3.1	4.5.1
Minor				
201	Evidence of poor workmanship	Level V	3.8	Visual

NOTES:

1. One sample should be selected for verification test at the beginning of each eight hour production.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Bridged header and ignition assembly	1 OF 1	12550793
			NEXT HIGHER ASSEMBLY: 12550991

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Spot charge missing	Level IV	3.2	Visual
102	Cracks or voids in spot charge	Level IV	3.2	Visual
Minor				
201	Evidence of poor workmanship	Level V	3.8	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Header assembly	1 OF 1	12550795
			NEXT HIGHER ASSEMBLY: 12550794

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Pull test of electrode	50-0-1	3.2	4.5.4
102	Outside diameter, max.	Level IV	3.2	Gage
103	Bead cracked	Level IV	3.2	Visual
104	Gold Plating	Note 1	3.2	Gage
Minor				
201	Length of electrode below surface of header	Level V	3.2	Gage
202	Thickness of header	Level V	3.2	Gage
203	Evidence of poor workmanship	Level V	3.8	Visual

NOTES:

1. Sampling and rejection shall be in accordance with MIL-G-45204.

AMSMC Form 1570b-E, 1 Jul 89

Replaces AMSMC Form 1570, 1 Feb 85, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7	Battery primer assembly	1 OF 2	12550991
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Special				
a	Cup cracked or split	100%	3.2	Visual
b	Explosive on exterior of cup	100%	3.2	Visual
Minor				
101	Diameter, Max.	Level IV	3.2	Gage
102	Length of body	Level IV	3.2	Gage
103	Sealant above noted surface, max.	Level IV	3.2	Gage
104	Sealant not present 360 degrees	Level IV	3.2	Visual
105	Assembly damaged or distorted	Level IV	3.2	Visual
106	Bridgewire continuity - resistance	100%	3.2	4.5.5
	Sensitivity and Output Test (70 degrees +/- 10 degrees F)	125		4.5.2
		Note 1		
107	No-fire test	125-1-2	3.4.1	4.5.2.1
	Electrical	125-1-2	3.4.2	4.5.2.3
	Mechanical (after electrical)	125-1-2	3.4.3	4.5.2.2
108	All-fire output test			

NOTES:
1. Samples from No-fire test to be used to perform All-fire tests below.

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CLASSIFICATION OF CHARACTERISTICS

MIL-P-70825A (AR)

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.7	Battery primer assembly	SHEET 2 OF 2	12550991

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY / INSPECTION METHOD REFERENCE
	Environmental Tests - Insulation resistance, thermal shock, vibration and function	40 Note 1	3.5.1	4.5.3.1
109	No-fire test			
	Electrical	40-0-1	3.4.1	4.5.2.1
	Mechanical	40-0-1	3.4.2	4.5.2.3
	All-fire test			
	Electrical, Hot	10-0-1	3.5.2.1	4.5.2.4.1
	Electrical, Cold	10-0-1	3.5.3.1	4.5.2.4.1
	Mechanical, Hot	10-0-1	3.5.2.2	4.5.2.4.2
	Mechanical, Cold	10-0-1	3.5.3.2	4.5.2.4.2
110	Waterproofness test	40	3.5.4	4.5.3.2
	All-fire test		3.4.3	4.5.3.2/4.5.2.2
	Electrical hot	10-0-1	3.5.2.1	4.5.2.4.1
	Elecrrical cold	10-0-1	3.5.3.1	4.5.2.4.1
	Mechanical hot	10-0-1	3.5.2.2	4.5.2.4.2
	Mechanical cold	10-0-1	3.5.3.2	4.5.2.4.2
Minor				
201	Length of electrode	Level V	3.2	Gage
202	Evidence of poor workmanship	Level V	3.8	Visual

NOTES:

1. Samples from No-fire test to be used to perform All-fire tests below.

AMSMC Form 1570b-E, 1 Jul 89

Rer ces AMSMC Form 1570, 1 Feb 85, which may no e used.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.8	Sleeve, header	1 OF 1	12550796
			NEXT HIGHER ASSEMBLY: 12550795

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Outside diameter	Level IV	3.2	Gage
102	Overall length	Level IV	3.2	Gage
Minor				
201	Workmanship	Level V	3.8	Visual

NOTES:

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection Tables.

4.4.4 Inspection equipment. The inspection equipment required to perform the inspections specified herein is identified in the "Inspection Method Reference" column of the First Article table or Classification of Characteristics listings starting with 4.4.2.1. Contractor inspection equipment designs shall be submitted for Government approval as specified in the contract. See section 6 of MIL-A-48078 and 6.3.

4.4.5 Function test equipment. The test fixture and procedure for subjecting the Battery Primer to No-fire and All-fire Output Tests, as evidenced by diameter expansion of witness sleeve shall be as specified on Drawings 12550754 and Figure 2.

4.5 Methods of inspection.

4.5.1 Moisture content of primer input mixture. The moisture content of the primer mixture shall be determined at the time of loading. A sample of 0.7 grams shall be selected. The sample shall be accurately weighed to the nearest milligram and dried to a constant weight at 60 degrees C ± 5 degrees C, or in a desiccator containing anhydrous calcium chloride. The loss in weight shall be calculated as the percentage of moisture as follows:

Percentage of moisture 100 (A-B)/A
Where:
A = The weight of the sample before drying
B = The weight of the sample after drying

4.5.2 Functioning tests-sensitivity and output. The Battery Primer samples selected from each inspection lot shall be assembled to the applicable test fixture (see 4.4.5.1) and subjected to a No-fire test followed by an All-fire output test.

4.5.2.1 No-fire test . The Battery Primer shall be conditioned at 70 degrees ± 10 degrees F for two (2) hours minimum (min.) prior to test. The Battery Primer shall be subjected to a constant 10 ma current for 10 seconds minimum as specified in 3.4.1. Any Primer which fails to comply with the applicable requirements as specified in 3.4.1 shall be classed defective (non-destructive test).

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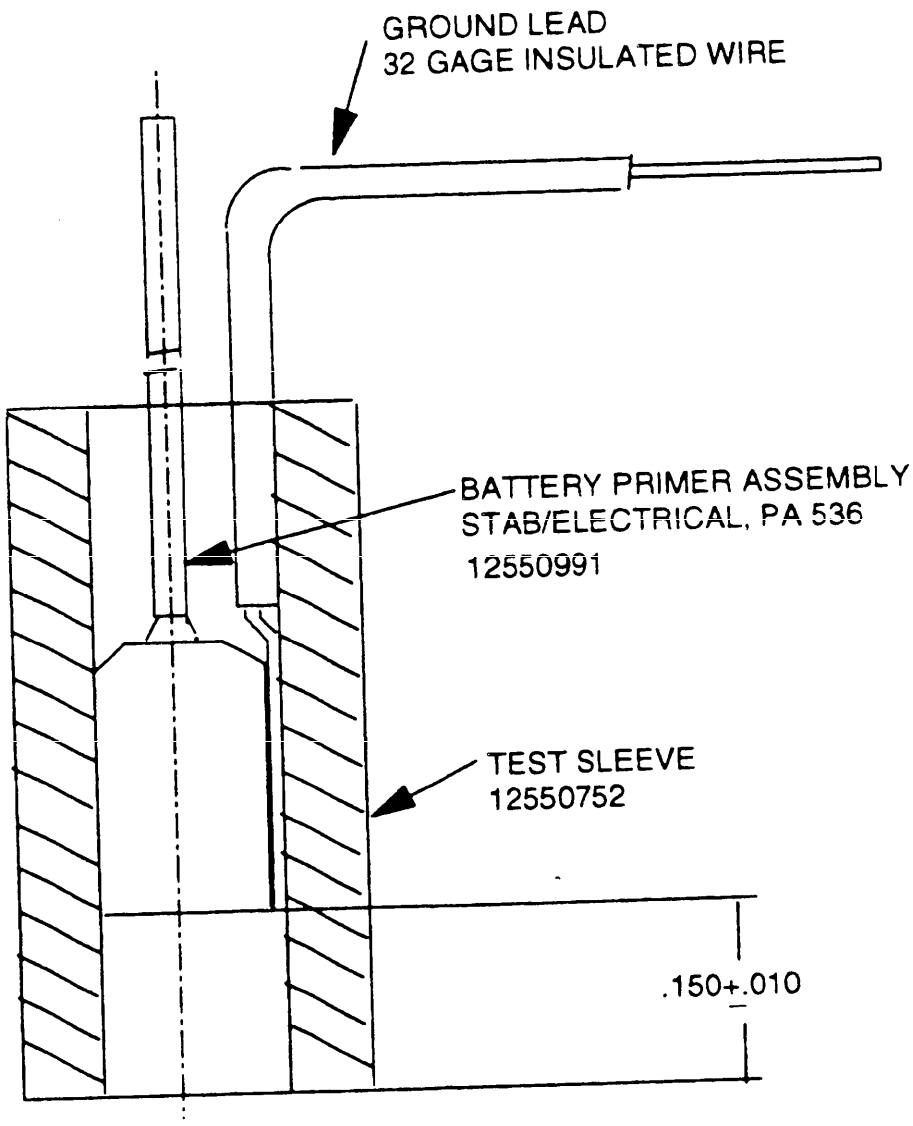


FIGURE 2. Output test sleeve assembly

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4.5.2.2 All-fire output test. Following the No-fire test of 4.5.2.1, the Battery Primer shall be assembled to the applicable test fixture (see 4.4.5.1) and fired by applying the energy from the circuit shown in Figure 1. Observation shall be made for failure to comply with requirements specified in 3.4.3. If any Battery Primer fails to meet the output requirements on the first attempt, it shall be classed defective (destructive test).

4.5.2.3 Mechanical no-fire. The Battery Primer assembly shall not function when dropped base down from a height of 5 feet minimum onto a flat steel plate. Assemblies subjected to this test shall not be returned to the lot. This test may be conducted at a temperature of 70 degrees F ± 10 degrees F.

4.5.2.4 All-fire function test.

4.5.2.4.1 Electrical. The Battery Primer shall be fired by applying the energy from the circuit shown in Figure 1. If any Battery Primer fails to function on the first attempt it shall be classed defective (Destructive Test).

4.5.2.4.2 Mechanical. The battery primer shall be fired by stabbing a steel firing pin into the stab sensitive surface of the Battery Primer with a maximum energy of 5 inch-ounces. The tip of the steel firing pin shall be 16 degrees -2 degrees right circular cone and a .007 ± .001 diameter flat. The test shall be performed using a Ball Drop Tester shown on Ordnance Corps Drawing 81-3-150 or equivalent. The firing pin shall be positioned within a 0.07 inch diameter area on the stab sensitive surface. If any Battery Primer fails to function on the first attempt, it shall be classed defective (Destructive Test).

4.5.3 Environmental tests.

4.5.3.1 Resistance, thermal shock, vibration and function. The Battery Primer sample shall be subjected to the required tests in the following sequence:

a. Condition for two (2) hours minimum at -65 degrees ± 5 degrees F.

b. Remove from cold chamber and within 20 seconds transfer to a hot chamber at 160 degrees ± 5 degrees F.

c. Condition for two (2) hours minimum at 160 ± 5 degrees F and 20% maximum relative humidity.

d. Remove from hot chamber.

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e. Vibrate for 15 minutes, minimum in a sweep covering a frequency range of 25-500-25 cycles per second, parallel to the transverse orthogonal axis, with base charge down and a 2 ± 0.2 g's peak amplitude.

f. Measure resistance in accordance with 4.5.5.

g. The Primers shall be assembled to test fixture (see 4.4.5) and tested as specified in 4.5.2.1 and 4.5.2.2. Observation shall be made to determine compliance with requirements of 3.4.1 and 3.5.1. Any Primer which fails to comply with the applicable requirement shall be classed defective (destructive test).

4.5.3.2 Waterproofness. Battery Primers from each lot shall be subjected to this test. The Battery Primers shall be immersed in water to a minimum depth of two (2) inches for 48 hours, minimum. The required water temperature of 70 degrees ± 10 degrees F shall be maintained for the immersion period. At the end of the immersion time, the Battery primers shall be removed from the water, assembled into a test fixture (see 4.4.5.1) and fired by applying the energy from the test circuit shown in Figure 1. Observation shall be made to determine compliance with the requirements of 3.4.4. If any Battery Primer fails to meet the requirements on the first attempt it shall be classed defective. If one or more defects occur, the lot shall be rejected (destructive test).

Note: An alternative method for testing seal integrity may be submitted by the contractor.

4.5.4 Pull test of electrode of header assembly (Dwg. 12550795). The header assembly shall be assembled in an approved fixture and the procedure of MIL-STD-202, Method 211, Test Condition A shall be used, with the exception that the force specified on the drawing shall be applied to the electrode for 10 seconds minimum. The force shall be a gradual, static axial load. Any header in which either the electrode moves or loosens shall be classed defective. This test is a destructive test.

4.5.5 Bridgewire continuity resistance of battery primer. Prior to shorting, the Battery Primer shall be connected to approved equipment and the bridgewire (electrode to cup) resistance measured to determine compliance with the applicable requirement specified on Drawing 12550991. Any primer which fails to comply with the applicable requirement shall be classed defective and removed from the lot. The measurement shall be made with a meter capable of measurement to the nearest 0.01 ohm. The current applied

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shall be limited to one (1) milliampere, maximum (non-destructive test).

5. PACKAGING

5.1 Unit Packaging.

5.1.1 Level A. Not Applicable.

5.1.2 Level B. Not Applicable.

5.1.3 Level C. Battery Primer shall be packaged in accordance with MIL-STD-1190 and the applicable section(s) of Code of Federal Regulation Title 49. In addition, each primer shall be electrically shorted during shipment and storage by means of a metallic connection between electrode and cup.

Unless other packaging is used to provide barrier protection, the following instructions apply.

Each intermediate fiberboard box, set-up box or comparable container, after blunting corners, shall be placed in a water and vapor proof bag manufactured in accordance with Type I, Class E of MIL-B-117. The bag shall be closed. All excess air shall be exhausted from within the bag and the bag then sealed in accordance with the manufacturer's instructions. The seal shall be located at the edge of the bag to provide for a minimum of two (2) subsequent seals, if required. The seal shall comply with the heat seal requirements of MIL-P-116.

5.2 Packing.

5.2.1 Level A. Not Applicable.

5.2.2 Level B. Not Applicable.

5.2.3 Level C. A suitable number of Battery Primers packaged as specified in 5.1.3 shall be packed in an outer box in accordance with MIL-STD-1190 and the applicable section(s) of Code of Federal Regulations, Title 49.

5.3 Marking.

5.3.1 Level A. Not Applicable.

5.3.2 Level B. Not Applicable.

5.3.3 Level C. Each inner container and intermediate

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fiberboard box, set-up box or comparable container shall be printed, stamped or labeled and each barrier bag printed or stamped as required by applicable section(s) of Code of Federal Regulations, Title 49, and in accordance with MIL-STD-129, and the following information:

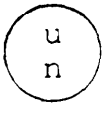
a. (Number) Battery, Primer, PA536: Lot Number.

b. All letters shall be upper case. Size of letters and numerals shall be a minimum of 1/4" high applied with black waterproof ink. Stencil ink shall conform to A-A-208.

c. Printing and labeling shall be commercial type. Labels shall be on white gummed paper.

d. Exterior boxes shall be marked in accordance with Drawing 8796522. The DOT Designation shall be in accordance with U.S. Army ARDEC HCSD statement. Apply appropriate label.

5.3.3.1 Performance oriented packaging marking. Apply POP markings on side of box opposite nomenclature:



4C1/Y*/S/**
USA/DOD/AYD

Note 1 -- *1/2 the weight of pack in pounds, plus 1.
Note 2 -- **Date packaged.

5.3.3.2 Unitization. Unitization shall be in accordance with MIL-STD-1190.

5.4 Packaging, packing and Marking for inert battery primer parts. Packaging, packing and marking for inert battery primer parts shall be in accordance with MIL-STD-1169 for all applicable levels of packaging.

5.5 Shipping. When Battery Primers from more than one lot are shipped at one time, each lot shall be kept separate and the division between lots clearly indicated to prevent mixing of lots in transit.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The Battery Primer, PA536 is intended for use in munition assemblies.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and, if required, the specific issue of individual documents referenced (see 2.1.1)

c. Requirements for submission of first article sample.

6.3 Submission of contractor inspection equipment designs for approval. Submit copies of designs as required to: Commander, U.S. Army ARDEC, ATTN: SMCAR-QAR-I, Picatinny Arsenal, New Jersey 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.4 Distribution of ammunition data cards, inspection and test results, process controls. Distribution of ammunition data cards, inspection and test results, processes controls shall be sent to:

Commander
ARDEC
ATTN: SMCAR-QAR-I
Picatinny Arsenal, New Jersey 07806-5000.

6.5 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development, and Engineering Center (ARDEC) may also include drawings prepared by, and identified as U.S Army Armament, Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is are now under the cognizance of ARDEC.

6.6 Submission of alternative quality conformance provisions. All contractor proposed alternative quality conformance provisions will be submitted to the Government for evaluation/approval as directed by the contracting activity.

6.7 Subject term (key word) listing.

Shock G, test
Waterproofness

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to

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e agent's authority with subsequent principal Substantiation of

the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1375-0005)

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-F-70825A (AR)	940602

3 DOCUMENT TITLE PRIMER, BATTERY, STAB/ELECTRIC PA536 PARTS FOR, AND LOADING, ASSEMBLING AND PACKING

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	f. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
U.S. ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial 201-724-6675	(2) AUTOVON DSN-880-6675
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 *Previous editions are obsolete.* 198/290

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